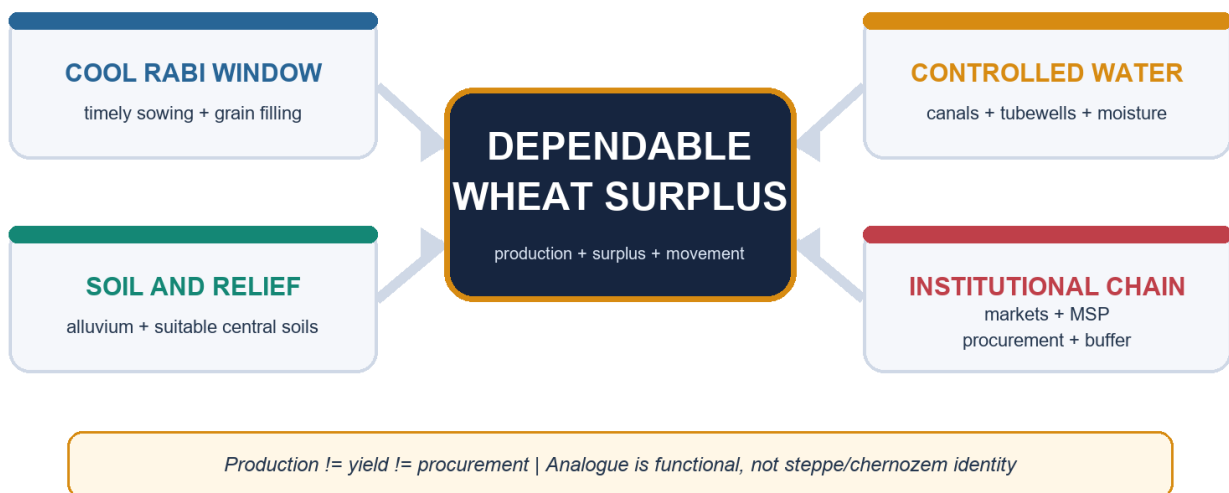


Geography 20 - India Wheat Granary - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

INDIA WHEAT GRANARY: A FUNCTIONAL SYSTEM



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

PYQ / PRACTICE / WORKBOOK START

PYQ disclosure: No direct standalone UPSC question titled "India's Wheat Granary" was verified in the checked local official-paper corpus. The questions below are genuinely adjacent verified questions on zero tillage, MSP, Green Revolution science, rice-wheat systems, irrigation, crop diversification, NFSA/PDS, buffer stocks and groundwater. They are not relabelled as direct wheat- granary PYQs. Mains papers have no objective key. For the 2020 Prelims question, the exact stem is verified but an official key was not located in the local corpus; the reasoned solution is prominently non-official.

PYQ 1 - UPSC Prelims 2020 Q83: Zero Tillage

Question: What is/are the advantage/advantages of zero tillage in agriculture?

1. Sowing of wheat is possible without burning the residue of previous crop.
2. Without the need for nursery of rice saplings, direct planting of paddy seeds in the wet soil is possible.
3. Carbon sequestration in the soil is possible.

Select the correct answer using the code given below:

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/CSP_2020_GS_Paper-1.pdf. Official key not held locally.

Reasoned solution - not an official key: D is the most defensible answer. Zero-till and residue-compatible seeders permit wheat sowing without field burning; direct-seeded rice can avoid the nursery-transplanting step under suitable systems; reduced disturbance and residue retention can support soil-carbon storage. [LIMIT] Outcomes depend on management, soil, residue and weed control. **Why this earns marks:** every statement is tested independently and key status is honest.

PYQ 2 - UPSC GS-III 2018 Q3: MSP and Low-Income Trap

Question: What do you mean by Minimum Support Price (MSP)? How will MSP rescue the farmers from the low income trap? (Answer in 150 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/GENERAL-STUDIES-PAPER-III.pdf.

Model solution:

- [CLAIM] MSP is an announced floor-price instrument intended to reduce downside price risk; it rescues income only where transmission and productivity conditions work.
- [EVIDENCE] CACP recommends and the Union government approves MSP; FCI/state procurement makes the signal credible for wheat in procurement-heavy regions.
- [ANALYSIS] A credible floor reduces distress-sale risk, supports investment in seed and inputs, improves credit confidence and can stabilise marketed surplus.
- [QUALIFICATION] Announcement without purchase access, quality acceptance, market competition or cost control may not raise realised income. Cereal concentration can also distort crops and water.
- [VERDICT] Combine MSP with wider markets, efficient procurement, storage, FPOs, risk management and region-suitable diversification.

Why this earns marks: It answers both definition and rescue mechanism, then qualifies coverage.

PYQ 3 - UPSC GS-III 2019 Q5: Visvesvaraya and Swaminathan

Question: How was India benefitted from the contributions of Sir M. Visvesvaraya and Dr. M. S. Swaminathan in the fields of water engineering and agricultural science respectively? (Answer in 150 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/QP-CSM19-GeneralStudies-III.pdf.

Model solution:

- [CLAIM] Visvesvaraya strengthened systematic water engineering; Swaminathan helped adapt modern crop science to Indian food-security needs.
- [EVIDENCE] Visvesvaraya is associated with automatic sluice gates and planned irrigation systems; Swaminathan's IARI leadership helped adapt Norin-10-derived semi-dwarf wheat with the wider Borlaug-IARI-state-extension effort.
- [ANALYSIS] Water control reduced production uncertainty, while fertiliser-responsive semi-dwarf wheat raised yield potential and supported national self-sufficiency.
- [QUALIFICATION] Neither achievement was individual or sufficient alone: dams need governance; seed required irrigation, fertiliser, credit, extension and procurement.
- [VERDICT] Their combined legacy is infrastructure plus adaptive science, now requiring ecological efficiency.

Why this earns marks: The two contributions remain distinct and are connected without hero-worship.

PYQ 4 - UPSC GS-III 2020 Q13: Rice-Wheat System

Question: What are the major factors responsible for making rice-wheat system a success? In spite of this success how has this system become bane in India? (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/Gen_St_P3.pdf.

Model solution:

- [CLAIM] The system succeeded because complementary technology and institutions stabilised two crops; it became a bane where the same incentives exceeded ecological capacity.
- [EVIDENCE] Punjab-Haryana-western UP combined canal/tubewell irrigation, semi-dwarf seed, fertiliser, mechanisation, electricity, mandis and paddy-wheat procurement.
- [ANALYSIS] Assured water and purchase enabled double cropping, high yield and central-pool surplus. Path dependence then narrowed rotation and encouraged pumping, nutrient imbalance, paddy-residue burning, pest/weed pressure and delayed wheat sowing.
- [QUALIFICATION] Wheat is generally less water-demanding than flooded paddy; the annual system and incentive structure, not wheat alone, drive the crisis.
- [VERDICT] Reform paddy incentives and groundwater governance, enable residue-compatible timely wheat, restore soils and build markets for region-suitable alternatives.

Why this earns marks: It gives success factors and negative consequences with a balanced causal verdict.

PYQ 5 - UPSC GS-III 2020 Q14: Water Storage and Irrigation

Question: Suggest measures to improve water storage and irrigation system to make its judicious use under depleting scenario. (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/Gen_St_P3.pdf.

Model solution:

- [CLAIM] Judicious irrigation needs source augmentation, distribution efficiency and demand governance together.
- [EVIDENCE] PMKSY components, watershed ridge-to-valley treatment, Atal Bhujal Yojana, canal command management and micro-irrigation provide named instruments.
- [ANALYSIS] Rehabilitate tanks and canals, harvest rain, recharge aquifers, schedule critical irrigations, improve field levelling, measure water budgets and align electricity/crop incentives.
- [QUALIFICATION] Drip or improved efficiency can produce rebound if farmers expand irrigated area; recharge cannot justify continued over-extraction.
- [VERDICT] Manage water at aquifer/canal-command scale with farmer institutions and crop planning, not through isolated structures.

Why this earns marks: Storage, conveyance, field use and incentives are all addressed.

PYQ 6 - UPSC GS-III 2021 Q13: NFSA and Hunger

Question: What are the salient features of the National Food Security Act, 2013? How has the Food Security Bill helped in eliminating hunger and malnutrition in India? (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/QP-CSM-21-GENSTUDIESPAPER-III-110122.pdf.

Model solution:

- [CLAIM] NFSA converted cereal access for eligible households into a legal-entitlement framework, but hunger and malnutrition require more than grain delivery.
- [EVIDENCE] Priority households, Antyodaya households, TPDS, maternity and child-related nutrition provisions, grievance architecture and central-pool procurement-distribution form the system.
- [ANALYSIS] Wheat and rice stocks stabilise household cereal access and protect consumption during income shocks; portability and digitisation can improve access for migrants.
- [QUALIFICATION] Exclusion, outdated coverage, biometric failure, diet quality, sanitation, health and intra-household allocation limit nutritional impact.
- [VERDICT] Preserve entitlements while updating coverage and integrating pulses, millets, maternal health and accountable delivery.

Why this earns marks: It distinguishes foodgrain access from complete nutrition security.

PYQ 7 - UPSC GS-III 2021 Q14: Crop Diversification

Question: What are the present challenges before crop diversification? How do emerging technologies provide an opportunity for crop diversification? (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/QP-CSM-21-GENSTUDIESPAPER-III-110122.pdf.

Model solution:

- [CLAIM] Diversification is constrained by coordinated market and risk failures, not merely farmer preference.

- [EVIDENCE] Rice-wheat regions possess specialised machinery, procurement, storage and skills; alternatives often lack buyers, processing and insurance depth.
- [ANALYSIS] Remote sensing and suitability maps improve crop matching; climate-resilient seed, micro-irrigation, advisories, precision input use, digital markets, traceability and processing lower production and transaction risk.
- [QUALIFICATION] Technology cannot create remunerative demand and can deepen debt or exclusion.
- [VERDICT] Build cluster-specific buyer, processing, extension, credit and transition-income packages before shifting area.

Why this earns marks: It links technology to each identified constraint rather than listing gadgets.

PYQ 8 - UPSC GS-III 2022 Q3: PDS Effectiveness

Question: What are the major challenges of Public Distribution System (PDS) in India? How can it be made effective and transparent? (Answer in 150 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/more_previous_papers/QP-CSM-22-GENERAL-STUDIES-PAPER-III-190922.pdf.

Model solution:

- [CLAIM] PDS challenges arise from beneficiary identification, stock movement, dealer incentives, authentication and grievance failures.
- [EVIDENCE] NFSA entitlements rely on procurement, central-pool movement, state allocation and fair-price shops; ONORC and ePoS are named transparency reforms.
- [ANALYSIS] Update rolls, publish allocation and stock data, ensure portability, audit transport and shops, maintain offline fallback and strengthen social audit/grievance redress.
- [QUALIFICATION] Digitisation can reduce diversion but exclude genuine users through connectivity, biometric or seeding failure.
- [VERDICT] Combine traceability with human fallback, independent audit and nutritional diversification.

Why this earns marks: It follows grain from procurement to last-mile access.

PYQ 9 - UPSC GS-III 2024 Q13: Irrigation Challenges

Question: What are the major challenges faced by Indian irrigation system in recent times? State the measures taken by the government for efficient irrigation management. (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/mains/03 UPSC 2024 Paper - III.pdf.

Model solution:

- [CLAIM] India's irrigation problem is simultaneous under-access, low efficiency and over-extraction.
- [EVIDENCE] Canal tail-end inequity, siltation, groundwater-energy incentives and regional crop concentration coexist; PMKSY, Per Drop More Crop, Har Khet Ko Pani and Atal Jal address different links.
- [ANALYSIS] Modernise canals, improve scheduling and field channels, scale suitable drip/sprinkler, budget aquifers, use advisories and reform crop/power incentives.
- [QUALIFICATION] Micro-irrigation is not universally suitable for wheat and does not ensure basin saving without extraction limits.
- [VERDICT] Shift from project completion to water-service, aquifer and crop-outcome monitoring.

Why this earns marks: The answer distinguishes access, delivery efficiency and sustainability.

PYQ 10 - UPSC GS-III 2024 Q14: Buffer Stocks

Question: Elucidate the importance of buffer stocks for stabilizing agricultural prices in India. What are the challenges associated with the storage of buffer stock? Discuss. (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/mains/03 UPSC 2024 Paper - III .pdf.

Model solution:

- [CLAIM] Buffer stocks stabilise availability and expectations only when procurement, storage and release are calibrated.
- [EVIDENCE] FCI/state agencies procure wheat and rice for the central pool, support NFSA and can release stocks through OMSS(D).
- [ANALYSIS] Purchase supports farm prices; stocks bridge seasons and emergencies; calibrated release augments supply. Storage faces moisture, pests, ageing, excess stock, carrying cost, location mismatch and movement constraints.
- [QUALIFICATION] Excess procurement can distort crops and water, while aggressive release can weaken farmer prices.
- [VERDICT] Use transparent dynamic norms, scientific storage, decentralised capacity, quality monitoring and rule-based release.

Why this earns marks: It answers both stabilisation and storage, then adds policy trade-offs.

PYQ 11 - UPSC GS-III 2025 Q13: Groundwater Depletion

Question: Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater? (Answer in 250 words)

Source status: VERIFIED PYQ - exact English stem checked in local official paper books/mains/UPSC Mains 2025 GS Paper 3 3.pdf.

Model solution:

- [CLAIM] Depletion results from an open-access aquifer interacting with cheap pumping, crop incentives, urban demand and weak recharge governance.
- [EVIDENCE] The north-west rice-wheat system illustrates power-procurement-crop lock-in; Atal Jal, Jal Shakti Abhiyan, watershed programmes and aquifer mapping are named responses.
- [ANALYSIS] Meter or proxy-monitor extraction, budget aquifers, recharge suitable zones, improve canal reliability, align crop purchase with water, promote precision irrigation and strengthen local user institutions.
- [QUALIFICATION] Recharge potential varies by geology; wheat is not equivalent to flooded paddy; efficiency can rebound into greater total use.
- [VERDICT] Govern aquifers as common resources and publish water-crop-energy outcomes.

Why this earns marks: It explains causal interaction, names measures and avoids blaming one crop.

Solved Original MCQ and Remedial Loops

Rotation audit: the 48 original keys below run continuously A -> B -> C -> D, repeated exactly twelve times. PYQ option order above is preserved separately and excluded from this rotation.

HIGH UPSC RELEVANCE

HIGH

Objective-key Disclosure - UPSC Prelims 2020 Q83:
Zero TillagePrelims
Geography +
Agriculture

NEWS TRIGGER

What is/are the advantage/advantages of zero tillage in agriculture? Statements cover residue-compatible wheat sowing, direct-seeded rice and soil carbon sequestration.

INTRO & ORIGIN

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: D - 1, 2 and 3. Confidence: High. The exact English stem was checked in the local official paper; a readable local official key was not available.

STATIC THEORY

- Residue-compatible zero-till seeders can establish wheat without field burning in suitable systems.
- Direct-seeded rice can avoid the nursery-transplanting step; reduced disturbance plus residue retention can support soil-carbon storage.
- Qualification: weed, residue, soil, moisture and machine management determine outcomes.
- Why this earns marks: every statement is evaluated independently and the inferred-key status is prominent.

MUST-KNOW FACTS

1. Verified PYQ; inferred key.
2. Zero tillage is one tool within conservation agriculture.

UPSC TRAPS

WRONG: Present D as an official local key.

CORRECT: Retain the inferred-answer label and confidence.

MAINS ANGLE

Use as a qualified conservation-agriculture example, not a universal yield claim.

STUDY LINK

Rice-wheat turnaround -> zero tillage -> residue -> soil carbon.

HIGH

Supplemental PYQ A - UPSC Prelims 2018 Q93: MSP
CropsPrelims
Geography +
Economy

NEWS TRIGGER

Consider the following: 1. Areca nut 2. Barley 3. Coffee 4. Finger millet 5. Groundnut 6. Sesamum 7. Turmeric. The Cabinet Committee on Economic Affairs has announced the Minimum Support Price for which of the above?

INTRO & ORIGIN

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: B - 2, 4, 5 and 6 only. Confidence: High. The exact English stem was checked in the local official paper; a readable local official key was not available.

KEY DATA

Option	Text
A	1, 2, 3 and 7 only
B	2, 4, 5 and 6 only
C	1, 3, 4, 5 and 6 only
D	1, 2, 3, 4, 5, 6 and 7

STATIC THEORY

- Barley, finger millet, groundnut and sesamum are MSP crops in the listed set; areca nut, coffee and turmeric are not in that CACP MSP list.

- Elimination: Remove options containing plantation/spice items areca nut, coffee or turmeric.

- Why this earns marks: every statement or crop is tested independently, and the missing-key status is explicit.

MUST-KNOW FACTS

1. PYQ status is verified; answer-key status is inferred.
2. Do not rewrite the original option order.

UPSC TRAPS

WRONG: Treat a coaching key as an official UPSC key.

CORRECT: Retain the inferred-answer label and confidence.

MAINS ANGLE

Use the concept in MSP, groundwater, irrigation and food-system answers.

STUDY LINK

Verified local UPSC paper -> concept elimination -> official static source.

HIGH

Supplemental PYQ B - UPSC Prelims 2019 Q79: FCI Economic Cost

Prelims
Geography +
Economy

NEWS TRIGGER

The economic cost of foodgrains to the Food Corporation of India is Minimum Support Price and bonus, if any, paid to the farmers plus:

INTRO & ORIGIN

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: C - procurement incidentals and distribution cost. Confidence: High. The exact English stem was checked in the local official paper; a readable local official key was not available.

KEY DATA

Option	Text
A	transportation cost only
B	interest cost only
C	procurement incidentals and distribution cost
D	procurement incidentals and charges for godowns

STATIC THEORY

- FCI economic cost combines pooled/procurement cost with distribution cost; procurement incidentals form part of the acquisition side.
- Elimination: Options A, B and D omit a major cost block or substitute one component for the full distribution cost.
- Why this earns marks: every statement or crop is tested independently, and the missing-key status is explicit.

MUST-KNOW FACTS

1. PYQ status is verified; answer-key status is inferred.
2. Do not rewrite the original option order.

UPSC TRAPS

WRONG: Treat a coaching key as an official UPSC key.
CORRECT: Retain the inferred-answer label and confidence.

MAINS ANGLE

Use the concept in MSP, groundwater, irrigation and food-system answers.

STUDY LINK

Verified local UPSC paper -> concept elimination -> official static source.

HIGH

Supplemental PYQ C - UPSC Prelims 2020 Q69: Unlimited MSP Procurement

Prelims
Geography +
Economy

NEWS TRIGGER

Consider the following statements: 1. For all cereals, pulses and oilseeds, procurement at MSP is unlimited in any State/UT. 2. For cereals and pulses, MSP is fixed at a level to which market price will never rise. Which is/are correct?

INTRO & ORIGIN

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: D - Neither 1 nor 2. Confidence: High. The exact English stem was checked in the local official paper; a readable local official key was not available.

KEY DATA

Option	Text
A	1 only
B	2 only
C	Both 1 and 2
D	Neither 1 nor 2

STATIC THEORY

- Statement 1 overstates operational procurement; commodity, quality, centres and scheme rules matter. Statement 2 misunderstands MSP: market prices can rise above it.
- Elimination: Both statements use absolute language that contradicts procurement practice and price formation.
- Why this earns marks: every statement or crop is tested independently, and the missing-key status is explicit.

MUST-KNOW FACTS

1. PYQ status is verified; answer-key status is inferred.
2. Do not rewrite the original option order.

UPSC TRAPS

WRONG: Treat a coaching key as an official UPSC key.
CORRECT: Retain the inferred-answer label and confidence.

MAINS ANGLE

Use the concept in MSP, groundwater, irrigation and food-system answers.

STUDY LINK

Verified local UPSC paper -> concept elimination -> official static source.

HIGH

Supplemental PYQ D - UPSC Prelims 2020 Q98: CGWA and Groundwater Irrigation

Prelims
Geography +
Economy

NEWS TRIGGER

Consider the following statements: 1. 36% of India's districts are classified as overexploited or critical by CGWA. 2. CGWA was formed under the Environment (Protection) Act. 3. India has the largest area under groundwater irrigation in the world. Which is/are correct?

INTRO & ORIGIN

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: B - 2 and 3 only. Confidence: High. The exact English stem was checked in the local official paper; a readable local official key was not available.

KEY DATA

Option	Text
A	1 only
B	2 and 3 only
C	2 only
D	1 and 3 only

STATIC THEORY

- Statement 1 confuses assessment units with districts and is therefore false. CGWA's legal basis and India's groundwater-irrigation scale make statements 2 and 3 correct.
- Elimination: Reject every option containing statement 1; retain the option combining 2 and 3.
- Why this earns marks: every statement or crop is tested independently, and the missing-key status is explicit.

MUST-KNOW FACTS

1. PYQ status is verified; answer-key status is inferred.
2. Do not rewrite the original option order.

UPSC TRAPS

WRONG: Treat a coaching key as an official UPSC key.
CORRECT: Retain the inferred-answer label and confidence.

MAINS ANGLE

Use the concept in MSP, groundwater, irrigation and food-system answers.

STUDY LINK

Verified local UPSC paper -> concept elimination -> official static source.

HIGH

Supplemental PYQ E - UPSC Prelims 2023 Q76: Global Groundwater Withdrawal

Prelims
Geography +
Economy

NEWS TRIGGER

Statement I: According to the UN World Water Development Report 2022, India extracts more than a quarter of the world's groundwater withdrawal each year. Statement II: India needs this share to satisfy drinking water and sanitation needs of almost 18% of world population. Which is correct?

INTRO & ORIGIN

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: C - Statement I correct, Statement II incorrect. Confidence: High. The exact English stem was checked in the local official paper; a readable local official key was not available.

KEY DATA

Option	Text
A	Both correct; II explains I
B	Both correct; II does not explain I
C	I is correct but II is incorrect
D	I is incorrect but II is correct

STATIC THEORY

- India's very large withdrawal is established, but agriculture is the dominant use; drinking water and sanitation alone do not explain the quarter-share.
- Elimination: The reason statement misattributes the sectoral driver, so it cannot be correct.
- Why this earns marks: every statement or crop is tested independently, and the missing-key status is explicit.

MUST-KNOW FACTS

1. PYQ status is verified; answer-key status is inferred.
2. Do not rewrite the original option order.

UPSC TRAPS

WRONG: Treat a coaching key as an official UPSC key.
CORRECT: Retain the inferred-answer label and confidence.

MAINS ANGLE

Use the concept in MSP, groundwater, irrigation and food-system answers.

STUDY LINK

Verified local UPSC paper -> concept elimination -> official static source.

HIGH

Supplemental PYQ F - UPSC GS-I 2024 Q14: Gangetic Groundwater and Food Security

GS-I | 15 marks |
250 words
Geography

NEWS TRIGGER

The groundwater potential of the Gangetic valley is on a serious decline. How may it affect the food security of India?

INTRO & ORIGIN

VERIFIED PYQ - exact English stem checked in the local official paper. The answer must trace aquifer decline through production risk, distribution and nutrition, while avoiding a single-crop explanation.

CLAIM -> EVIDENCE -> ANALYSIS -> QUALIFICATION -> VERDICT**1. Claim**

Aquifer decline converts a high-surplus plain into a reliability risk.

3. Analysis

higher pumping cost, unequal access, delayed irrigation, yield/quality volatility and procurement risk.

5. Verdict

manage crop-water-energy incentives without abrupt foodgrain withdrawal.

2. Evidence

CGWB 2025 national assessment plus rice-wheat dependence and dense irrigation.

4. Qualification

effects vary across upper, middle and lower plains; drainage and surface-water access differ.

Reading order: left to right, then top to bottom.

STATIC THEORY

- [CLAIM] Declining groundwater threatens the timing and reliability of irrigation in one of India's largest cereal-surplus regions.
- [EVIDENCE] The Gangetic plain combines wheat/rice intensity, productive aquifers, large populations and major market-procurement networks; CGWB's 2025 national assessment confirms the need to measure recharge and extraction rather than assume abundance.
- [ANALYSIS] Falling water tables raise energy and capital costs, deepen inequality between deep-well owners and small/tenant farmers, and can delay irrigation at CRI or reproductive stages. Yield and grain quality become more variable; procurement and buffer replenishment can face spatial shocks.
- [ANALYSIS] Food security effects extend from availability to access: higher costs and price volatility affect consumers, while excessive cereal lock-in can crowd out pulses and nutrition diversity.
- [QUALIFICATION] The valley is heterogeneous. Some canal commands face waterlogging/salinity; eastern areas may have groundwater potential but weak energy, drainage, seed or market access. Wheat is also less water-intensive than conventionally flooded paddy.
- [VERDICT] Combine aquifer accounting, conjunctive use, reliable but better-priced energy support, paddy-side diversification, timely wheat, efficient application, recharge/drainage, procurement reform and income safeguards.
- Why this earns marks: it answers the food-security pathway, names evidence, differentiates the valley, qualifies crop blame and ends with a feasible policy package.

MUST-KNOW FACTS

1. Food security includes availability, access, stability and utilisation.
2. Water-table depth is an equity and energy issue, not only a hydrological statistic.

UPSC TRAPS

WRONG: Blame wheat alone for Gangetic depletion.

CORRECT: Analyse the rotation, energy-water incentives and spatial variation.

MAINS ANGLE

A small schematic map plus the aquifer-to-food-security chain earns geography value addition.

STUDY LINK

Groundwater -> rice-wheat system -> procurement and price stability -> nutrition.

HIGH

Supplemental MCQ Set - Official Zones and Agronomy (S1-S4)

Prelims
Geography

NEWS TRIGGER

Authored supplement; strict key sequence continues A -> B -> C -> D.

STATIC THEORY

- Reference: the full official names are Northern Hills Zone, North Western Plains Zone, North Eastern Plains Zone, Central Zone, Peninsular Zone and Southern Hills Zone.

- S1. Which is the official six-zone sequence used by IIWBR? A. NHZ, NWPZ, NEPZ, CZ, PZ, SHZ | B. Himalayan, Desert, Delta, Plateau only | C. North, South, East, West only | D. Procurement zones. Answer: A. It is the six-zone agro-climatic research framework.

- S2. Which statement is most accurate? A. India cultivates only bread wheat | B. Bread wheat dominates, with durum and dicoccum in smaller niches | C. Durum dominates Punjab | D. Dicoccum is a procurement category. Answer: B. The guide-era orientation is roughly 95:4:1.

- S3. Under severe irrigation scarcity, which stage receives highest priority in the IIWBR guide? A. post-harvest | B. dough only | C. crown-root initiation | D. storage. Answer: C. CRI establishes roots and tiller potential.

- S4. Which claim is unsafe? A. stage sensitivity matters | B. local advisory matters | C. soil and variety alter scheduling | D. 21/45/65/85/105/120 DAS is universal for every field. Answer: D. It is an indicative guide.

MUST-KNOW FACTS

1. S1-S4 key rotation: A, B, C, D.
2. Each option was checked against the IIWBR guide and status caveats.

UPSC TRAPS

WRONG: Treat zone or DAS guidance as a legal/universal boundary.

CORRECT: Retain research and local-advisory context.

MAINS ANGLE

Use zones and stages to regionalise any wheat-climate answer.

STUDY LINK

IIWBR pocket guide -> zoning -> agronomy -> adaptation.

HIGH

Supplemental MCQ Set - 2026 Status Firewall (S5-S8)

Prelims Current
Geography +
Economy

NEWS TRIGGER

Authored supplement; strict key sequence continues A -> B -> C -> D.

STATIC THEORY

- S5. What did 106.36 LMT denote in the accessed official table? A. Central Pool wheat opening stock on 1 April 2026 | B. final 2026 procurement | C. buffer norm | D. sowing area. Answer: A.

- S6. The 46,000 machines, 910 CHCs and 141 supply-chain projects in the June 2026 CRM release are best read as: A. audited outcomes | B. targets/status under the provision | C. farmer counts | D. stock norms. Answer: B.

- S7. Which evidence sequence is methodologically sound? A. forecast -> certain loss -> trend | B. one anomaly -> climate attribution | C. observed weather -> crop assessment -> trend -> projection | D. estimate -> final without revision. Answer: C.

- S8. Which statement is defensible as of the retrieval date? A. 303 LMT was final procurement | B. 334.17 lakh ha was harvested area | C. 120.657 Mt was final output | D. the accessed primary sources did not establish a final RMS 2026-27 procurement outcome. Answer: D.

MUST-KNOW FACTS

1. S5-S8 key rotation: A, B, C, D.
2. Total authored sequence is now 56 questions: fourteen complete A-B-C-D cycles.

UPSC TRAPS

WRONG: Drop the status word when revising current data.

CORRECT: Write date + unit + scale + status every time.

MAINS ANGLE

A status firewall prevents data fabrication and overclaiming.

STUDY LINK

Official releases -> evidence status -> exam-safe current linkage.

End of Register Notes - UPSC Agent / Copilot CLI

MCQ 1

Which formulation most accurately describes India's wheat granary?

- A. A functional irrigated and institutional analogue of world wheat granaries, not a true steppe B. A chernozem belt identical to the Ukrainian steppe C. A statutory zone notified by FCI D. A rainfed belt confined to Punjab

Answer: A. India's granary role rests on winter crop suitability, irrigation, technology, marketed surplus and procurement; climatic and pedological identity with the steppe is false.

MCQ 2

Which soil comparison is correct?

- A. Punjab alluvium and chernozem have identical origins B. Punjab-Haryana wheat is mainly linked with alluvium; central Indian wheat may also use black soils C. Every black soil in India is a temperate grassland soil D. Wheat can grow only on chernozem

Answer: B. The northern core is alluvial, while parts of central India use regur or mixed soils; neither should be relabelled chernozem.

MCQ 3

Why is one rigid India-wide wheat calendar unsafe?

- A. Wheat is grown only in hill regions B. Wheat is a perennial crop C. preceding crop, variety, irrigation, latitude and altitude shift farm windows D. MSP changes the biological crop duration

Answer: C. Regional agroclimate and farm sequence determine actual sowing and harvest windows.

MCQ 4

Terminal heat most directly threatens wheat by:

- A. creating chernozem B. increasing winter chill indefinitely C. extending grain filling D. accelerating development and shortening grain filling

Answer: D. Late-season heat can compress the period in which grain accumulates dry matter.

MCQ 5

In the official 2024-25 state table, which listed state had the largest wheat production?

- A. Uttar Pradesh B. Punjab C. Haryana D. Bihar

Answer: A. Uttar Pradesh produced 35.65 million tonnes in the cited DES table.

MCQ 6

In the same dated table, which listed state had the highest wheat yield?

- A. Madhya Pradesh B. Punjab C. Uttar Pradesh D. Rajasthan

Answer: B. Punjab's reported yield was 5,123 kg per hectare, illustrating why yield and total production ranks differ.

MCQ 7

Which sequence correctly moves from farm output to public purchase?

- A. procurement -> production -> yield -> area B. MSP -> stock -> rainfall -> seed C. production -> marketable/marketed surplus -> eligible arrival -> procurement D. buffer stock -> yield -> marketable surplus -> sowing

Answer: C. Government procurement is a filtered part of marketed output, not a synonym for production.

MCQ 8

Which statement about marketed surplus is most accurate?

- A. It is always identical to procurement B. It equals yield minus area C. It includes only grain stored by FCI D. It is the quantity actually sold after farm retention and other uses

Answer: D. Marketed surplus includes public and private sales and differs from marketable surplus in timing and farmer retention decisions.

MCQ 9

The 2024-25 production-procurement comparison most strongly shows that:

- A. production leadership and procurement leadership can differ sharply B. Punjab produced more wheat than Uttar Pradesh C. public purchase equals total output in every state D. yield has no role in production

Answer: A. UP's much larger production coexisted with far lower procurement than Punjab.

MCQ 10

Why can Punjab-Haryana be called granary anchors despite not leading every output rank?

- A. They have a statutory monopoly on wheat B. high yield, marketed surplus, purchase centres and logistics combine there C. all their wheat is exported D. their wheat is rainfed

Answer: B. Granary function includes dependable surplus extraction and movement, not output alone.

MCQ 11

Which physical combination best supports north Indian wheat?

- A. steep slopes, laterite and summer cyclones B. mangrove mud and tidal irrigation C. level alluvial plains, cool winter and assured irrigation D. permanent snow and tundra soil

Answer: C. Relief, soil, winter weather and water control form the physical platform.

MCQ 12

Which statement about western disturbances is correct?

- A. They are a branch of the southwest monsoon B. More events always improve yield C. They affect only Himalayan tourism D. moderate timely rain may help, while hail or rain near maturity can damage wheat

Answer: D. Their agricultural effect is conditional on intensity, track and crop stage.

MCQ 13

What best explains Madhya Pradesh's growing wheat role?

- A. later diffusion of suitable varieties, irrigation, markets and procurement over diverse soils B. conversion into a true Eurasian steppe C. complete absence of climate risk D. exclusive dependence on chernozem

Answer: A. MP demonstrates dynamic diffusion beyond the original north-west core.

MCQ 14

Which distinction between regur and chernozem is correct?

- A. Both are simply names for Punjab alluvium B. regur is basalt-linked Indian black soil; chernozem is a temperate grassland soil C. chernozem is a coastal saline soil D. regur cannot support any cereal

Answer: B. Similar colour or crop use does not establish identical soil genesis.

MCQ 15

Which was indispensable to the Green Revolution wheat package?

- A. seed without water or fertiliser B. procurement without production technology C. complementarity among semi-dwarf seed, water, inputs, credit, extension and markets D. replacement of every traditional crop in India

Answer: C. The yield response depended on a package whose components reinforced one another.

MCQ 16

Why did early Green Revolution gains concentrate spatially?

- A. all Indian regions had identical preconditions B. wheat biology prohibited eastern cultivation C. procurement was illegal outside Punjab D. assured water, infrastructure, credit and markets were unevenly distributed

Answer: D. Adoption followed the geography of enabling preconditions.

MCQ 17

What is path dependence in granary formation?

- A. past procurement and infrastructure lower current transaction costs and reinforce the same region B. a river permanently fixes crop choice without institutions C. a crop cannot ever move to another state D. yield automatically falls after one season

Answer: A. Existing mandis, storage, machinery and administrative capacity make continued specialisation more attractive.

MCQ 18

Which statement best captures eastern Indo-Gangetic constraints?

- A. Alluvial fertility guarantees Punjab-like procurement B. delayed rice harvest, water control, small surplus and market access can delay wheat intensification C. wheat cannot grow east of Delhi D. eastern India lacks winter

Answer: B. Physical fertility must be joined with timing, institutions and controlled water.

MCQ 19

Which statement about wheat irrigation is best supported?

- A. High-output Indian wheat is primarily a rainfed steppe system B. Irrigation coverage proves groundwater is sustainable C. official data show wheat is overwhelmingly irrigated, while sources and sustainability vary D. canals are the sole wheat-water source

Answer: C. The cited all-India irrigation share is high, but canal, groundwater and residual moisture combinations differ.

MCQ 20

Which is the safest conclusion about wheat and paddy water use?

- A. Both require identical continuous ponding B. Wheat has no water footprint C. Paddy never affects groundwater D. wheat generally needs less water than flooded paddy, but still matters within the annual nexus

Answer: D. The groundwater crisis is a system outcome, not permission to ignore wheat irrigation.

MCQ 21

The water-energy-crop nexus refers to:

- A. pumping incentives, irrigation security and crop choice reinforcing groundwater extraction
- B. photosynthesis alone
- C. export freight charges
- D. western disturbances creating electricity

Answer: A. Power pricing and pump access alter both private costs and aquifer pressure.

MCQ 22

Why may plot-level irrigation efficiency fail to reduce basin extraction?

- A. efficient systems cannot deliver water
- B. saved water may support area expansion or more intensive cropping
- C. aquifers refill instantly
- D. crop choice never changes water demand

Answer: B. This is the efficiency-rebound problem.

MCQ 23

Conservation agriculture is best defined by:

- A. burning all residue before sowing
- B. deep tillage every season
- C. minimum disturbance, residue management and crop rotation
- D. replacing irrigation with rainfall

Answer: C. Zero tillage is one component, not the entire conservation system.

MCQ 24

How can residue-compatible zero tillage support climate resilience?

- A. by creating winter rainfall
- B. by eliminating all weeds
- C. by converting regur into alluvium
- D. by reducing turnaround delay and helping wheat avoid later terminal heat

Answer: D. Timely establishment can move sensitive grain filling away from hotter late-season conditions.

MCQ 25

What is the most direct agronomic consequence of terminal heat?

- A. compressed grain filling and lighter grain
- B. creation of more tillers after maturity
- C. longer vernalisation
- D. higher soil humus by definition

Answer: A. Heat accelerates late crop development and can reduce the time available for grain weight accumulation.

MCQ 26

Which is an avoidance strategy rather than a genetic-tolerance strategy?

- A. breeding a heat-tolerant variety
- B. suitable early/timely sowing to shift grain filling
- C. changing alluvium into black soil
- D. increasing storage after harvest

Answer: B. Calendar adjustment aims to avoid exposure; tolerant varieties aim to perform better under exposure.

MCQ 27

Which statement follows the climate-evidence ladder?

- A. One hailstorm proves a century-long trend
- B. A projection is an observed event
- C. event, anomaly, trend, attribution and projection require separate evidence
- D. national output reveals every district's loss

Answer: C. Each evidence category answers a different question and requires its own period and method.

MCQ 28

Why can national output remain high despite local heat damage?

- A. heat cannot damage wheat B. procurement creates grain biologically C. state tables are never revised D. larger area, timely sowing or better varieties can offset local yield loss in the aggregate

Answer: D. Aggregate production can conceal spatially concentrated losses and compensatory gains.

MCQ 29

What does public wheat procurement primarily do in spatial terms?

- A. converts selected surplus arrivals into stocks for national movement and distribution B. measures all household wheat consumption C. determines winter temperature D. replaces private markets everywhere

Answer: A. Procurement links surplus geographies to the central pool.

MCQ 30

Which statement about the RMS 2026-27 figure of 303 LMT is correct?

- A. It is verified achieved procurement as of 19 August 2026 B. It is an official procurement target, not an achieved total C. It is India's wheat production D. It is a farmer count

Answer: B. The source is a March 2026 planning target and must retain that status.

MCQ 31

Which statement correctly interprets the RMS 2025-26 farmer count?

- A. It covers every wheat farmer in India B. It proves 3.54 crore wheat farmers received MSP C. it reports 2,513,143 farmers associated with completed public wheat procurement D. it is the number of FCI warehouses

Answer: C. The official count applies to procured sellers and is about 25.13 lakh.

MCQ 32

Why are quality norms central to procurement geography?

- A. they determine monsoon onset B. they replace MSP C. they make storage unnecessary D. weather, harvest moisture and handling affect whether grain is accepted and stored safely

Answer: D. Procurement is a quality-filtered transaction, not an automatic purchase of every grain lot.

MCQ 33

What is a core function of buffer stocks?

- A. supporting distribution, emergencies and calibrated market stabilisation B. guaranteeing that every crop receives procurement C. measuring soil fertility D. setting the sowing date

Answer: A. Public stocks connect procurement with welfare, stability and emergency response.

MCQ 34

Which statement distinguishes OMSS(D) from TPDS?

- A. Both are identical entitlement channels B. OMSS(D) releases stocks into open-market channels; TPDS delivers entitlements C. TPDS is a crop-breeding programme D. OMSS(D) fixes rainfall

Answer: B. The two releases use public grain for different policy purposes.

MCQ 35

Why can excess stocks become problematic?

- A. grain cannot be stored at all B. they eliminate carrying costs C. storage, quality, movement and fiscal costs rise, while crop incentives may distort D. they automatically diversify diets

Answer: C. Security stocks require calibration; quantity alone is not an unqualified good.

MCQ 36

Which statement about MSP transmission is correct?

- A. notification guarantees a sale at every village B. MSP has no ex-ante effect on crop decisions C. private prices can never exceed MSP D. realised support depends on access, quality, operations and alternative buyers

Answer: D. The price signal passes through several spatial and institutional filters.

Remedial MCQ 37

A student writes, "Punjab is India's largest wheat producer, so it is the granary." What is the best correction?

- A. Granary status should be explained through yield, surplus and procurement; dated output data put UP ahead B. Punjab grows no wheat C. production is irrelevant to granary analysis D. only soil colour matters

Answer: A. The correction preserves Punjab's granary role while separating current production rank from procurement and yield.

Remedial MCQ 38

Which statement repairs the "India is a steppe wheat belt" error?

- A. India has no economic wheat regions B. India is a functional granary analogue built on monsoonal winter cropping, alluvium and irrigation C. Punjab is part of the Eurasian steppe D. chernozem covers the Ganga plain

Answer: B. Functional similarity must be paired with physical and institutional difference.

Remedial MCQ 39

A map shows wheat across UP, MP, Punjab and Haryana. What cannot be inferred from that map alone?

- A. wheat occurs in several states B. regional extent exists C. the ranking of government procurement D. India has a Rabi season

Answer: C. A crop-occurrence or output map is not a procurement map.

Remedial MCQ 40

Which is the best qualification to "western disturbances help wheat"?

- A. They are tropical cyclones B. They occur only in summer C. Their rain is never useful D. benefit depends on timing and intensity; hail or maturity-stage rain can cause damage

Answer: D. Stage-specific direction of effect is the essential repair.

Remedial MCQ 41

Which statement avoids blaming wheat for all north-west groundwater depletion?

- A. the annual rice-wheat-power-procurement system and hydrogeology jointly determine stress B. wheat uses no irrigation C. paddy uses no water D. procurement cannot affect crop choice

Answer: A. The system explanation is more accurate than single-crop causation.

Remedial MCQ 42

What must accompany a diversification recommendation?

- A. only a slogan to reduce wheat B. agroclimatic fit plus buyers, processing, risk cover and transition support C. a ban on every cereal D. a national one-crop replacement

Answer: B. Farmers need a viable alternative income system, not just a different seed.

Remedial MCQ 43

Which statement about the official 2025-26 production number is correct?

- A. It is a final state-wise table B. It proves no district suffered heat stress C. it is a national Third Advance Estimate of 120.657 million tonnes D. it is the wheat procurement target

Answer: C. Status, scale and date must remain attached to the figure.

Remedial MCQ 44

Why was the claimed 3.54 crore wheat-farmer coverage excluded?

- A. farmer counts are never official B. no procurement occurs in India C. the package avoids all numbers D. the checked completed official procurement count was 2,513,143 and no official confirmation supported 3.54 crore

Answer: D. The package uses the verified scoped count and rejects an unsupported aggregate.

Remedial MCQ 45

What is the best answer order for a wheat-granary Mains question?

- A. map/definition -> physical base -> technology/institutions -> distinctions -> stresses -> reform B. schemes -> conclusion -> climate -> question C. data list without thesis D. global comparison only

Answer: A. The order moves from spatial foundation to causal system and qualified reform.

Remedial MCQ 46

Which metric best tests whether a granary is sustainable?

- A. production alone B. yield and income together with aquifer, soil, climate and procurement outcomes C. MSP announcement alone D. number of tractors alone

Answer: B. Sustainability is a multi-outcome test, not a tonnes-only claim.

Remedial MCQ 47

Which statement correctly handles Madhya Pradesh?

- A. It has replaced every north-west granary function B. It is climatically identical to Punjab C. it is a major and growing production/procurement region with a different physical pathway D. it grows wheat only on chernozem

Answer: C. MP updates the geography without erasing regional difference.

Remedial MCQ 48

What is the strongest final verdict?

- A. Remove wheat from India B. Restore a prairie climate in Punjab C. Expand procurement everywhere regardless of surplus D. retain wheat where suitable, reform water-crop incentives, diversify viably and monitor dated outcomes

Answer: D. A regionally differentiated systems strategy balances food security and resilience.

Original Solved Mains Practice

Original Mains 1 - 10 marks: Functional Analogue

Question: Why is the Indo-Gangetic wheat belt a functional rather than climatic analogue of the world's temperate wheat granaries? (150 words)

Model solution:

- [CLAIM] The belt performs the surplus-grain function of the steppe/prairie but reaches it through a monsoonal, irrigated and institutionally dense pathway.
- [EVIDENCE] Punjab-Haryana-western UP combine cool Rabi weather, level alluvial plains, canal and groundwater irrigation, semi-dwarf wheat, mechanisation and procurement. Eurasian/North American belts are associated with temperate continental climates, chernozem/prairie soils and extensive commercial farms.
- [ANALYSIS] Both systems convert plains and crop suitability into large tradable surplus, yet irrigation and central-pool procurement substitute in India for the steppe's climatic and farm- structure advantages.
- [QUALIFICATION] MP's expansion further shows that India's granary is dynamic and not confined to one north-west climate-soil unit.
- [VERDICT] Use "functional analogue" to compare economic role while rejecting environmental identity.

Why this earns marks: It defines the comparison, names both similarity and difference and ends with a precise geographical verdict.

Original Mains 2 - 10 marks: Four Different Rankings

Question: Distinguish wheat production, yield, marketed surplus and procurement, using India's regional pattern. (150 words)

Model solution:

- [CLAIM] The four measures answer different questions: total output, land productivity, grain sold and public purchase.
- [EVIDENCE] DES 2024-25 data show UP at 35.65 MT production, Punjab at 5,123 kg/ha yield, and public procurement of 12.457 MT in Punjab versus 0.931 MT in UP.
- [ANALYSIS] UP's large area generates output; Punjab's technology raises yield; household retention and private sales affect marketed surplus; centre density, quality and agencies shape procurement.
- [QUALIFICATION] Procurement/output is not a marketed-surplus ratio and state orders are dated.
- [VERDICT] A granary answer should overlay crop, market and state-purchase geographies rather than declare one universal ranking.

Why this earns marks: One official contrast demonstrates the conceptual distinction efficiently.

Original Mains 3 - 10 marks: Weather at the Right Stage

Question: Explain why the effect of weather on wheat must be analysed by crop stage rather than seasonal average. (150 words)

Model solution:

- [CLAIM] The same weather variable can help or harm wheat depending on phenological stage.
- [EVIDENCE] Moderate western-disturbance rain can support vegetative moisture, while hail or rain near maturity causes lodging and quality loss; MoA&FW's April 2026 assessment linked February heat with shorter grain filling.
- [ANALYSIS] Establishment needs workable moisture, flowering is sensitive to frost/heat, grain

filling needs adequate duration, and maturity benefits from dry weather.

- [QUALIFICATION] A national seasonal mean can conceal local timing, duration, variety and irrigation differences.

- [VERDICT] Answers should map hazard -> crop stage -> mechanism -> outcome.

Why this earns marks: It replaces descriptive weather with a causal crop-stage framework.

Original Mains 4 - 15 marks: Formation of the North-West Granary

Question: Analyse the physical and human factors that transformed Punjab-Haryana-western Uttar Pradesh into India's wheat-granary core. (250 words)

Model solution:

- [CLAIM] The core was created when a favourable winter plain was connected to a complementary technology-market-state package.

- [EVIDENCE] The physical platform comprised level Indo-Gangetic alluvium, cool Rabi weather, residual moisture, canals and productive aquifers. The Green Revolution added Norin-10-derived semi-dwarf wheat, fertiliser, credit, extension, power and mechanisation; FCI/state procurement linked surplus to the central pool.

- [ANALYSIS] Level relief enabled machines and irrigation; controlled water made input-responsive varieties reliable; semi-dwarf form reduced lodging; rapid machines protected the rice-wheat turnaround; MSP/procurement reduced marketing uncertainty. Repeated surplus then attracted mandis, storage and political-administrative capacity, producing path dependence.

- [QUALIFICATION] Physical conditions were not sufficient, and success was unequal across farmers and regions. The system later accumulated groundwater, nutrient and residue stress.

- [VERDICT] Punjab-Haryana became a granary through applied geography, but its next phase requires paddy-side water reform, timely residue-compatible wheat and soil restoration.

Why this earns marks: It explains causality, feedback and the second-generation qualification.

Original Mains 5 - 15 marks: Madhya Pradesh and a Changing Granary

Question: Madhya Pradesh's growing wheat role complicates the old Punjab-centred granary map. Discuss. (250 words)

Model solution:

- [CLAIM] MP shows diffusion of wheat surplus beyond the original north-west core without making Indian wheat geography uniform.

- [EVIDENCE] DES 2024-25 reports MP at 24.51 MT, second among the listed states, with 4.839 MT public procurement; Punjab produced 17.99 MT but procured 12.457 MT.

- [ANALYSIS] MP's large area, suitable cool-season windows, moisture-retentive black and other soils, irrigation, improved varieties and procurement expansion raised output. Yet Punjab retains higher yield and much denser procurement relative to production.

- [QUALIFICATION] "Black soil" must not be relabelled chernozem, and MP contains varied soils and farm systems. One year's ranks cannot be frozen.

- [VERDICT] Replace the single-core map with a network: north-west high-yield/procurement anchors, UP production scale and MP's expanded production-procurement role.

Why this earns marks: It uses dated comparative evidence and updates, rather than discards, the granary concept.

Original Mains 6 - 15 marks: Procurement Geography and Food Security

Question: How does concentrated wheat procurement convert regional surplus into national food security, and what problems does this concentration create? (250 words)

Model solution:

- [CLAIM] Concentrated procurement is the territorial bridge between farm surplus and national cereal entitlements, but it also concentrates incentives and costs.
- [EVIDENCE] FCI/state agencies procure FAQ wheat, store it in the central pool, move it across states and issue it for NFSA/TPDS or OMSS(D). RMS 2025-26 procurement was officially 300.35 LMT; RMS 2026-27's 303 LMT is only a target.
- [ANALYSIS] Dense centres lower transaction costs and stabilise farm expectations in Punjab, Haryana and MP. Stocks bridge seasons, welfare demand and emergencies. Concentration, however, reinforces cereal specialisation, regional inequity, transport cost, storage pressure and water- crop lock-in.
- [QUALIFICATION] Wider procurement is not automatically efficient; regions need genuine surplus, quality, logistics and fiscal justification, while private markets remain legitimate.
- [VERDICT] Diversify procurement and diets selectively, modernise storage and use transparent stock/release rules without weakening dependable wheat supply.

Why this earns marks: The answer follows the grain chain and balances security with distortion.

Original Mains 7 - 20 marks: Sustainability of the Rice-Wheat Core

Question: Critically examine the sustainability crisis of the Punjab-Haryana rice-wheat system and propose a sequenced transition. (300 words)

Model solution:

- [CLAIM] The crisis is not a failure of wheat alone but the outcome of a successful double-crop system whose incentives now exceed local water, soil and atmospheric capacity.
- [EVIDENCE] Assured paddy-wheat procurement, cheap pumping, canal/tubewell irrigation, fertiliser response and specialised machinery produced high surplus. The same system is associated with aquifer decline, nutrient imbalance, short paddy-wheat turnaround and residue burning. The 2020 UPSC rice-wheat PYQ itself frames success turning into bane.
- [ANALYSIS] Paddy creates much of the north-west groundwater pressure; wheat adds annual pumping demand. Delayed paddy harvest compresses wheat sowing, encourages rapid residue clearance and increases terminal-heat exposure. Narrow rotation weakens biological diversity and reinforces input dependence.
- [SEQUENCED TRANSITION] First, preserve timely wheat through residue-compatible seeders and advisories. Second, reform paddy area and power-water incentives by aquifer zone. Third, guarantee buyers, processing and risk cover for suitable maize, pulses, oilseeds, fodder or horticulture. Fourth, restore balanced nutrients, residue use and rotation. Fifth, monitor water, income, grain quality and inclusion.
- [QUALIFICATION] Abrupt withdrawal of procurement would transfer ecological reform costs to farmers and risk central-pool stability; alternatives differ by district.
- [VERDICT] Protect the comparatively suitable Rabi wheat while restructuring the water-intensive Kharif and incentive system around it.

Why this earns marks: It identifies system causation, protects food security and sequences a just regional transition.

Original Mains 8 - 20 marks: Climate-Resilient National Wheat Strategy

Question: Design a regionally differentiated strategy for a climate-resilient, resource- efficient and food-secure Indian wheat economy. (300 words)

Model solution:

- [CLAIM] National resilience requires different interventions for the over-intensified north-west, large-output UP, expanding MP, eastern plains and marginal dry/hill systems.
- [EVIDENCE] The 2025-26 Third Advance Estimate places wheat at 120.657 MT; MoA&FW's April 2026 assessment records terminal-heat pressure, early-sowing adaptation and local rain/hail damage; DES 2024-25 separates state production, yield and procurement.
- [ANALYSIS] NORTH-WEST: protect timely wheat, reduce paddy groundwater load, rebalance nutrients and manage residue. UP: differentiate western aquifer stress from eastern turnaround, drainage and market constraints. MP: match irrigation and varieties to black/alluvial subregions and strengthen quality logistics. RAJASTHAN: expand only within sustainable water budgets. BIHAR/EAST: improve seed, drainage, machinery services and competitive buyers without copying north-west input intensity. HILLS: use local calendars and frost management.
- [SYSTEM MEASURES] Strengthen heat-tolerant/location-approved varieties, stage-specific IMD advisories, conservation agriculture, scientific storage, transparent stock release and insurance. Align procurement with quality and genuine surplus; enable viable diversification.
- [QUALIFICATION] Early sowing and resilient varieties are not universal prescriptions; advance estimates and targets must retain status; private trade remains part of the system.
- [VERDICT] Judge success by stable grain, farmer income, aquifer and soil outcomes, procurement efficiency and equitable regional access.

Why this earns marks: It is region-specific, evidence-led, institutionally complete and measurable.

Advanced Refinements and Examiner Cautions

Visual 55 - Weak-to-strong answer upgrades

Weak statement	Examiner-grade replacement
Punjab is India's largest wheat producer	Punjab is a high-yield and procurement anchor; UP led dated 2024-25 production
India has a steppe wheat belt	India has a functional granary analogue under different climate, soils and institutions
Wheat caused groundwater depletion	rice-wheat-power-procurement incentives interact with local aquifers; paddy is generally more water-intensive
Western disturbances are good for wheat	moderate timely rain can help; intense/maturity-stage rain or hail can damage
MSP guarantees farmer income	MSP is a signal whose effect depends on purchase access, quality and costs
More buffer stock is safer	stocks support security but excess raises carrying, quality and distortion costs
Diversify away from wheat	retain suitable wheat and build viable alternatives by agroclimate, buyer and water budget
Climate change cut output	identify event, stage, local loss, adaptation, aggregate offset and evidence status

Visual 56 - Data-status firewall

2024-25 state table = dated production/yield/procurement pattern

2025-26 120.657 MT	= Third Advance Estimate
RMS 2025-26 300.35 LMT	= completed procurement total
RMS 2026-27 303 LMT	= target
RMS 2026-27 Rs 2,585/q	= notified MSP

Never merge these into one undated "current figure."

Advanced synthesis

- [REFINEMENT] Granary is a network property: production nodes, procurement nodes, storage nodes and consumption destinations.
 - [REFINEMENT] Yield gaps can be closed only within water, soil and climate limits.
 - [REFINEMENT] Procurement geography is both an efficiency outcome and a distributive political-economy choice.
 - [REFINEMENT] Central-pool security should be measured through quantity, quality, location, turnover and delivery reliability.
 - [REFINEMENT] Heat adaptation must be stage-specific and paired with the paddy-to-wheat turnaround.
 - [REFINEMENT] Soil carbon, residue and zero tillage belong to one system but are not synonyms.
 - [REFINEMENT] MP's rise strengthens a polycentric granary; it does not validate chernozem analogy.
 - [REFINEMENT] Regional diversification must account for price covariance, processing capacity, labour demand and transition risk.
 - [REFINEMENT] A record national crop can coexist with local disaster, quality loss and unequal farmer outcomes.
 - [REFINEMENT] Every current number should carry year, unit, geographic scale and status.
-